

The Influence of Discourse Context on Children's Use of Conversational Devices

Cynthia Boo and Letitia Naigles

1. Introduction

Conversational devices, such as backchannels, linguistic alignment, and discourse markers, are useful in organizing discourse, ensuring mutual understanding between speakers, and maintaining social relations. Adults have been documented to use these devices differently depending on the context (e.g., Dideriksen et al., 2023; Spaniol et al., 2024). Our research extends this area of research to school-aged children and investigates the role of context on the use of these conversational devices. Further, we extend the study to consider the relationship between these devices and social relations, namely friendship quality.

In verbal exchanges, an addressee can explicitly show they are paying attention and/or agree with the speaker by using verbal signals, or backchannels (Yngve, 1970). While backchannels are typically produced while someone else is speaking (as seen in (1), with < > indicating overlapped speech), they are not considered to be a form of interruption, as they do not allow the addressee to take the conversational floor away from the speaker (Carletta et al., 1997):

- (1) Child 1: We could flip this piece to create a < ramp>.
Child 2: < mhm>.

Instead, they are brief shows of interest, understanding, and/or agreement with what the speaker is saying (Yngve, 1970). In fact, researchers have also referred to backchannels as rapport markers, as they often reflect the relationship status between speakers (e.g., Tickle-Degnen & Rosenthal, 1990). In a task where adults were asked to pass a verbal story down a chain of people, and then subsequently asked to rate how much they enjoyed their interaction, Rifai and colleagues (2022) reported that fewer instances of backchanneling corresponded with lower ratings of interactional enjoyment, suggesting that the presence of backchannels

* Cynthia Boo, Department of Psychology, New York University (previously University of Connecticut), cynthia.boo@nyu.edu. Letitia R. Naigles, Department of Psychological Sciences, University of Connecticut, letitia.naigles@uconn.edu. This research was partially funded by a National Institute on Deafness and Other Communication Disorders grant to L. Naigles (R01DC016665). We thank the children and their families who participated, UConn KIDS, and the undergraduate research assistants who transcribed the audio files.

contributes something enjoyable to social interactions. However, much of the literature on backchannels has focused on adults, particularly strangers, engaging with one another in a lab-based context. There is some research that suggests children can interpret and produce verbal backchannels early in development (i.e., preschool period; Peterson, 1990). Whether or not children backchannel in discourse with a peer, namely a friend, and whether backchannels contribute to social relations for school-aged children, remain relatively unexplored.

Similar to backchanneling, linguistic alignment also contributes to discourse organization and mutual understanding. Linguistic alignment refers to the tendency for speakers to reuse one another's words (lexical alignment), parts of speech (syntactic alignment) and/or topic of discussion (semantic alignment) (Brennan & Clark, 1996; Garrod & Pickering, 2009). Alignment can be quantified in terms of rate (i.e., proportion of conversational turns in which alignment occurred) and level (i.e., degree of alignment when alignment occurred). Some researchers have argued that linguistic alignment is an explicit mechanism whereby individuals reuse their interlocutor's linguistic material in a deliberate and targeted way to signal mutual understanding and even the status and/or quality of the speakers' relationship (Brennan & Clark, 1996). This argument aligns with theories stemming from social psychology, such as the coordination-rapport hypothesis (Tickle-Degnen & Rosenthal, 1987), which posits that relationship satisfaction and rapport are positively correlated with coordinated interaction patterns, including linguistic coordination. In fact, Ireland and colleagues (2011) reported that speed daters were more likely to be a match (i.e., choose one another for a future date) if there was greater lexical alignment, particularly of function words (i.e., personal pronouns, articles, conjunctions, prepositions, etc.), during their brief conversations. However, much of the research surrounding children's linguistic alignment patterns has been in the context of adult-child discourse. In these studies, children generally align less to adults than adults to children. For example, across two investigations using the same sample of caregiver-child interactions, children lexically aligned at lower rates to adults ($M_{\text{children}} = 0.23$; Fusaroli et al., 2023a) than adults to children ($M_{\text{adults}} = 0.38$; Fusaroli et al., 2023b), suggesting a possible asymmetry in the discourse context where the adult leads the discourse. How children linguistically align in discourse with a peer, namely a friend, remains relatively unexplored.

Like backchannels and linguistic alignment, discourse markers also guide conversations and shape discourse (Schiffrin, 1987). Examples of discourse markers, according to the seminal work conducted by Schiffrin (1987) include *and*, *because*, *but*, *like*, *I mean*, *now*, *so*, *then*, *well*, and *you know*. These markers are not limited to simply connecting utterances: they are involved in managing social relations (Brinton, 2010; Schiffrin, 1987). For example, the discourse marker *well* can be used to signal an upcoming disagreement (Jucker, 1991). In (2), child two clearly disagrees with child one, but instead of outwardly disagreeing with them (e.g., *No, the ramp is really for the jumping point, not the end.*), which may lead to a direct conflict, child two prefaces their contribution with *well* to soften and mitigate the disagreement. Without *well*, an explicit

disagreement may result in a larger conflict and possibly a less positive interaction and relationship.

(2) Child 1: We should use this ramp as the end point.

Child 2: *Well*, the ramp is really for the jumping point, not the end.

Despite this clear interpersonal function of particular discourse markers, few studies have investigated whether the use of discourse markers contributes to relationship quality. For instance, does using this interpersonal conversational device make a relationship more positive? Further, studies have shown that discourse markers are acquired early in development. For example, Sprott (1992) reported that children as young as three use *well* during disputes with a peer; Odato (2013) reported that children as young as four use *like* during play sessions with their friends and siblings; Skarabela and colleagues (2023) reported that children as young as seven understand that *but* can mark both simple and complex contrasts. Therefore, given the robust literature suggesting that children are capable of using a diverse range of discourse markers by the time they are school-aged, the current study focuses on children and their use of discourse markers with their friend to answer the question of whether or not discourse markers contribute anything to friendship quality.

In adult-to-adult discourse, speakers often adjust their uses of these conversational devices depending on the context. For example, Dideriksen and colleagues (2023) elicited speech from 40 dyads of Danish-speaking adults to investigate how speakers adjust their uses of conversational devices, such as backchannels and linguistic alignment, according to contextual demands. Each dyad participated in two spontaneously occurring conversations (SOCs), where they answered open-ended conversation prompts, and two task-oriented conversations (TOCs), where pairs engaged in joint problem-solving tasks that required participants to collaborate and communicate to solve the tasks. One of the TOCs revolved around a computerized game, referred to as the Aliens game, where dyads were asked to converse with one another while they made decisions whether aliens presented on the screen were friendly, or not, and if they had the gems the participants needed to gain points. The other TOC revolved around a map task where the goal is for one person to provide directions and the other person to follow them. All instances of Danish backchannels (e.g., *mm* [*mm*], *åh* [*oh*]) were tallied, and linguistic alignment was calculated. The authors reported that adults produced more backchannels and higher rates of lexical and syntactic alignment, as well as higher levels of semantic alignment in SOCs, and higher levels of lexical and syntactic alignment in TOCs. According to the authors, these devices play unique roles depending on the context: backchannels can help maintain social relations via displays of interest while one speaker holds the conversational floor in SOCs, where the discourse content is driven by the speaker, whereas higher levels of lexical and syntactic alignment may be required for precision in goal-based TOCs. As such, it is important to consider the context when scrutinizing people's use of conversational devices.

While context is important for adults’ usage of conversational devices, it remains unclear whether children are sensitive to the same contextual demands. Although studies have found that children are sensitive to the situational context in ways that affect their language production (e.g., Hoff 2010), these studies often focus on younger children and their lexical choices, rather than the conversational devices that are used frequently by older school-aged children. Further, while these conversational devices of interest contribute to social relations (e.g., softening disagreements, Jucker, 1991; interactional enjoyment, Rifai et al., 2022), few studies have empirically assessed whether their uses are related to relationship quality. Given our sample of school-aged children, the current study aims to address these gaps by comparing how school-aged children use these conversational devices across different contexts and whether their uses are related to friendship quality.

2. Methods

2.1. Participants

Table 1 presents demographic information of the participants. Thirty English-speaking children between the age of 8 and 14 years nested in 15 friendship dyads participated in the current study. Six of the dyads were girl-girl pairings; seven of the dyads were boy-boy pairings; and two of the dyads were girl-boy pairings. Dyads of friends were non-familial and within 1-2 years of age from one another. The majority of the children have been friends for more than half of their lives.

2.2. Standardized Measures

2.2.1. Clinical Evaluations of Language Fundamentals—5th Edition

The Expressive Language Index (ELI) from the Clinical Evaluation of Language Fundamentals—5th Edition (CELF-5; Semel, Wiig, & Secord, 2013) was used as a measure of children’s overall expressive language abilities. The ELI was derived using scaled scores from the Word Classes, Formulated Sentences, and Recalling Sentences subtests (for the 8-year-olds) and the Formulated Sentences, Recalling Sentences, and Sentence Assembly subtests (for the 9-to-14-year-olds). On average, children had age-appropriate expressive language abilities (standard scores above 86; Table 1).

Table 1. Demographic information of participants

	Age in years	Length of friendship*	ELI
<i>M (SD)</i>	9.7 (1.2)	0.6 (0.3)	102.5 (12.6)
Range	8 – 14	0.1 – 1.0	83 - 122

Table 1. Demographic information of participants (continued)

Dyad #	Participant #	Age in years	Length of friendship*	ELI
1⁺	1	10	0.3	91
	2	10	0.3	114
2⁺	3	10	0.5	106
	4	10	0.5	83
3[^]	5	9	0.4	95
	6	8	0.5	98
4⁺	7	11	0.4	122
	8	10	0.4	102
5[^]	9	13	0.1	106
	10	14	0.1	98
6⁺	11	8	0.2	83
	12	9	0.2	83
7[^]	13	10	0.5	118
	14	10	0.5	102
8⁺	15	8	0.3	90
	16	8	0.3	88
9[#]	17	10	0.9	116
	18	9	1.0	114
10⁺	19	8	1.0	90
	20	8	1.0	108
11⁺	21	10	0.5	100
	22	11	0.5	89
12[^]	23	10	0.4	120
	24	10	0.4	91
13[#]	25	10	0.8	102
	26	10	0.8	120
14[^]	27	10	1.0	116
	28	10	1.0	114
15[^]	29	10	0.9	95
	30	9	1.0	120

* Length of friendship as a proportion of child's life; ⁺ = boy-boy pairing; [^] = girl-girl pairing; [#] = boy-girl pairing

2.2.2. Network of Relationships Inventory—Relationship Quality Version

The Network of Relationships Inventory—Relationship Quality Version (NRI-RQV Buhrmester & Furman, 2008) is a 30-item questionnaire that assesses children's self-perceptions about their friendship quality with a friend. The NRI-RQV assesses five positive features (i.e., companionship, disclosure, emotional support, approval, and satisfaction) and five negative features (i.e., conflict, criticism, pressure, exclusion, and dominance). Children were asked to rate items

on a Likert-type scale ranging from 1 (“almost never”) to 5 (“almost always”). Scales were created by averaging the three items that make up a particular friendship feature. Two composite friendship features (closeness and discord) were computed by calculating the mean of the positive scales and the negative scales, respectively. Table 2 presents the means and standard deviations for the entire sample on each friendship feature.

Table 2. Means and standard deviations of each friendship quality feature derived from the NRI

Friendship Feature	<i>M (SD)</i>	Range
Companionship	3.37 (0.86)	1.67 – 4.67
Disclosure	2.66 (1.13)	1.00 – 5.00
Satisfaction	4.42 (0.76)	2.33 – 5.00
Emotional Support	2.88 (1.06)	1.00 – 4.67
Approval	3.37 (0.67)	1.67 – 4.67
Closeness	3.34 (0.47)	1.67 – 4.60
Pressure	1.81 (0.77)	1.00 – 3.67
Conflict	1.58 (0.55)	1.00 – 3.00
Criticism	1.16 (0.26)	1.00 – 2.00
Dominance	2.27 (0.57)	1.00 – 3.00
Exclusion	1.53 (0.47)	1.00 – 2.67
Discord	1.67 (0.29)	1.13 – 2.27

NRI ratings are on a scale of 1 to 5.

2.3. Language Sampling Activities

2.3.1. Task-Oriented Conversational Tasks

We utilized two task-oriented conversational (TOC) tasks that have been validated for the age range we were interested in. The first TOC task was an alien categorization game, where each dyad played a computerized game in which they pretended they were space explorers (Fusaroli et al., 2012; Tylén et al., 2020). Each dyad was told that their spaceship ran out of fuel on a planet inhabited by aliens, and that some of the aliens had the gems the participants needed to fuel up their ship. The aliens differed from one another on five characteristics: color (green/blue), eyes (on stalks/not on stalks), arms (up/down), spots (spots/no spots), and legs. Based on preset characteristics, some aliens were friendly but did not have gems, so the children should wave and move on; some aliens were friendly and had gems, so the children should ask for a gem; some aliens were unfriendly and had no gems, so the children should run away from the alien; and some aliens were unfriendly but had gems, so the children should steal from the alien. Across 32 trials, the children had to converse with one another and make joint decisions. Dyads were awarded ten points for every correct answer and deducted two points for every incorrect answer.

The second TOC task was a marbles construction game, which involved conversation between children as they jointly constructed a mutually agreed-upon

design using assorted pieces and connectors (Bauminger et al., 2008). Children were instructed to use every piece to create a single design that would allow them to successfully roll marbles down the final construction. Unlike the alien categorization game, there was no “correct answer” associated with the marble construction game; instead, children were asked to decide on a design together, which could be changed at any point.

2.3.2. Spontaneously Occurring Conversation

Two periods of spontaneously occurring conversational (SOC) language samples were collected. The experimenter disengaged with the dyad under the guise of setting up for the next task for five minutes each. During this time, the dyads were told they could chat as they normally would. Age-appropriate items (e.g., magazines, Rubik’s cubes, tabletop tic-tac-toe, etc.) were provided to spur conversation.

2.4. Procedure

Parents and/or legal guardians of all participants provided written consent prior to participation, and all children provided oral and written assent. The entire protocol was administered in a quiet laboratory room or in a quiet room in one of the children’s homes. Six out of the 15 dyads completed the protocol in a quiet laboratory room; the remaining nine dyads completed the protocol in one of the children’s homes. The order of the TOC games was counterbalanced across dyads. The experimenter remained in the room throughout the protocol. After completing the activities with their friend, each child completed the CELF. A video camera was placed in the corner of the room to record the audiovisuals of the entire procedure. To maintain motivation and reduce fatigue, breaks were provided as needed.

2.5. Language Coding

The video recordings were transcribed by undergraduate research assistants in Datavyu (Datavyu Team, 2014). The video recordings were transcribed following typical CHAT conventions (MacWhinney, 2000). All utterances were segmented based on a full second, or longer, break. All transcripts were checked for reliability by a second transcriber, and any disagreements were discussed until both transcribers were in agreement.

Verbal backchannels were identified via utterance-by-utterance hand coding. First, all instances of *ah*, *aw*, *aye*, *mm*, *mhm*, *oh*, *okay*, *oo*, *nice*, and *yeah/yep/yes* were identified. This list of backchannels was derived from previous literature (see Carletta et al., 1997; Spaniol et al., 2024; Yngve, 1970). Following the strict criteria outlined in Carletta and colleagues (1997), all utterances containing these backchannels were initially included. Then, the following were subsequently excluded: (1) turn-initial backchannels (where a backchannel directly preceded a

more substantial utterance by the same speaker, such as “*mhm, and we should put this here.*”) and (2) answers to a question or bid posed by the speaker (e.g., “*mhm*” as an affirmative response to the speaker’s question). The remaining instances were counted as backchannels. Raw number of backchannel tokens was tallied, and proportion of backchannel tokens per utterance was computed.

Discourse markers were also identified via utterance-by-utterance hand coding. Following Schifffrin’s (1987) original list of discourse markers, the following were identified in each transcript: *and, because, but, like, I mean, now, so, then, well, and you know*. Given that discourse markers are syntactically optional, secondary checks were performed by removing the discourse marker from the utterance to determine whether the meaning of the utterance changed without the presence of the discourse marker. If the meaning did change, the identified discourse marker was not counted as a discourse marker; otherwise, it was counted as a discourse marker. Raw number of discourse marker tokens was tallied, and proportion of discourse marker tokens per utterance was computed.

Measures of lexical, syntactic, and semantic alignment rate and level were extracted using the open-source Python library Analyzing Linguistic Interactions with Generalizable techNiques (ALIGN; Duran, Paxton, & Fusaroli, 2019). For lexical and syntactic alignment, each interlocutor’s contiguous sequence of tokens (for lexical) and part-of-speech tags (for syntactic) are segmented into short chunks of increasing sizes, or n-grams. ALIGN then generates frequency vectors composed of the number of times unique n-grams occur for each interlocutor, within each turn. These vectors are used to create a similarity score which is calculated by assessing the angle between the two vectors (as a single turn-by-turn exchange). For semantic alignment, content words from each speaker’s turn are transformed into a high-dimensional semantic representation. The resulting speaker vector is compared to their conversational partner vector to obtain semantic alignment values. It is important to note that, in most cases, lexical alignment on its own will result in syntactic and semantic alignment (Fusaroli et al., 2023). Therefore, we adjusted the measures of syntactic and semantic alignment by statistically controlling for lexical alignment to identify their unique contributions over and above lexical alignment, as was previously done in Fusaroli et al. (2023). Average alignment rates and levels for each task were calculated per child.

2.6. Analysis Plan

Descriptive statistics were first performed to summarize the overall language patterns of all thirty children during the entire protocol (i.e., all language produced during the Aliens game, Marbles game, and both unstructured conversations combined). To account for differences in verbosity, analyses were conducted on proportion of backchannel and discourse marker tokens out of all utterances and alignment rates out of all conversational turns.

To test whether these conversational devices differed by activity (i.e., TOC versus SOC), we built Bayesian multilevel logistic regression models using the

brms package in R (Bürkner, 2017). The presence or absence of each language measure produced by a child nested in their respective dyad was predicted by activity type. Initial models were built by combining the language produced during the aliens and marbles tasks into a single TOC context and the two periods of spontaneously occurring conversational language samples into a single SOC context. For each model, weakly informative priors were used in combination with a gaussian distribution. Markov chain Monte Carlo (MCMC) estimation algorithms were applied, with 4 Markov chains and 2,000 iterations per chain (1,000 iterations for the warmup). All models converged, as indicated by the R-hat statistic values (i.e., $R\text{-hat} \leq 1.01$).

To determine whether these conversational devices were related to children's perception of friendship quality, data were analyzed using Bayesian multilevel models with children nested in their respective friendship dyads. These analyses were also conducted in R using the *brms* package (Bürkner, 2017). In our initial models, the conversational devices were treated as "predictors" to determine how much variance each conversational device independently contributed to the various measures of friendship quality, which served as the outcome measures of interest. These models included the following covariates: children's chronological age, length of friendship as a proportion out of each child's life, and overall expressive language abilities (i.e., ELI standard scores).

3. Results

3.1. Descriptive Summary of Children's Language During Tasks Combined

On average, children produced roughly 270 utterances and 410 conversational turns throughout the entire protocol. Roughly 5% of utterances contained a backchannel token, and roughly 34% of utterances contained a discourse marker token. In terms of lexical alignment rate, children reused their friend's words in approximately 33% of their conversational turns. They reused their friend's syntactic items (beyond lexical alignment) in approximately 13% of their conversational turns. Isolating conversational turns containing alignment (i.e., alignment levels), children reused substantial parts of their friend's lexical items (20%) and semantic content (50%) but substantially fewer sets of syntactic elements (8%).

3.2. Effect of context on children's uses of conversational devices

Figure 1 presents the effect of context on the conversational devices. Each ridge represents the density of frequency values observed for each individual device across the two types of contexts. Devices with peaks toward the left were less frequently used, while peaks to the right indicate greater frequency. The overlap between ridges highlights similarities between the two contexts, whereas distinct peaks suggest differences in device usage across contexts. Overall, we found higher rates and levels of lexical alignment in the context of TOCs than SOC. Conversely, we found greater uses of backchannels and discourse markers,

as well as higher rates of syntactic alignment and levels of semantic alignment, in SOC than TOCs. Syntactic alignment level did not significantly differ across contexts.

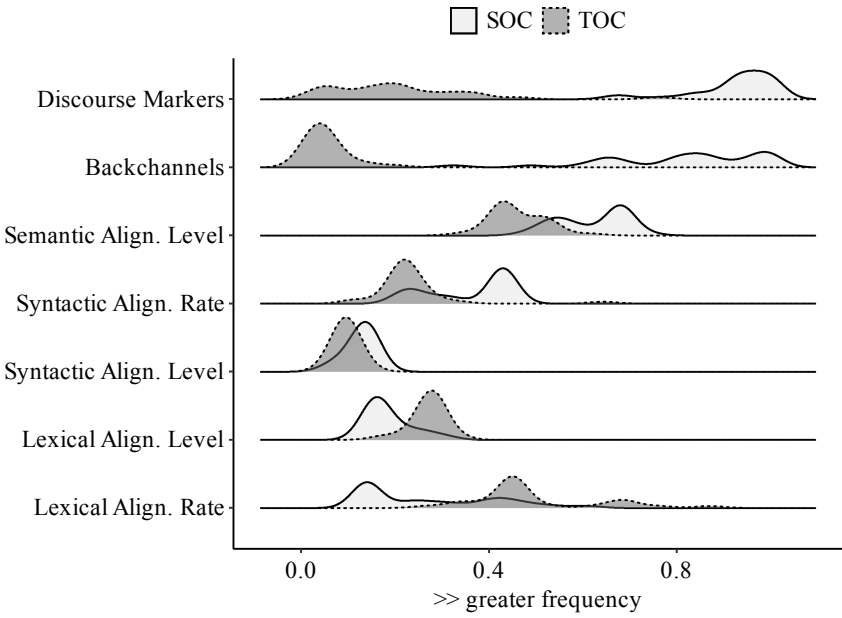


Figure 1. Most conversational devices varied by context

3.3. Relationship between conversational devices and time it took to complete games

Because the two TOC tasks were untimed, we performed exploratory analyses to determine the relationship between children’s use of conversational devices and the time it took to complete the games. Overall, there were no significant correlations between children’s use of backchannels or discourse markers and time. There were, however, significant correlations between lexical alignment rate ($r = -0.647$) and level ($r = -0.436$) and time, $ps < 0.05$ (Figure 2).

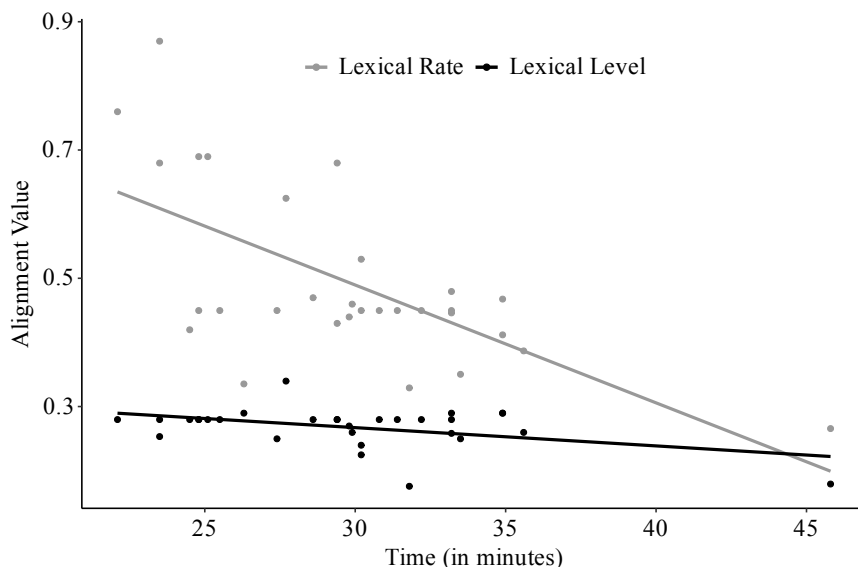


Figure 2. Greater lexical alignment rate and level were related to shorter completion times for TOCs

3.4. Relationship between conversational devices and friendship quality

Overall, there were no significant interaction effects between the conversational devices and the other child's perception of friendship (e.g., there was no significant effect of child's use of backchannels on their friend's perception of satisfaction, or vice versa). Therefore, the following results present the relationships between the conversational devices and children's own perceptions of their friendship (i.e., child 1's language and child 1's perception of their friendship; child 2's language and child 2's perception of their friendship). There were no significant effects of discourse markers, lexical alignment, or syntactic alignment on children's perceptions of their friendship quality. However, there was a significant effect of backchannels: children's use of backchannels was positively associated with their ratings of companionship ($\beta = 0.39$, $SE = 0.96$, 95% CI [1.76, 2.06]) and closeness ($\beta = 0.74$, $SE = 0.54$, 95% CI [0.43, 1.91]). Similarly, there was a positive association between children's semantic alignment levels and their ratings of companionship ($\beta = 0.25$, $SE = 0.95$, 95% CI [1.73, 2.05]), disclosure ($\beta = 0.20$, $SE = 0.98$, 95% CI [1.91, 2.92]), emotional support ($\beta = 0.19$, $SE = 0.92$, 95% CI [1.82, 2.13]), satisfaction ($\beta = 0.30$, $SE = 0.93$, 95% CI [1.53, 2.60]), and closeness ($\beta = 0.49$, $SE = 0.94$, 95% CI [1.30, 2.30]). Conversely, there was a negative association between children's semantic alignment levels and their ratings of conflict ($\beta = -0.33$, $SE = 0.96$, 95% CI [1.35, 1.79]), criticism ($\beta = -0.25$, $SE = 0.97$, 95% CI [1.35, 1.79]), and discord ($\beta = -0.41$, $SE = 0.94$, 95% CI [1.44, 2.26]).

4. Discussion

The current study aimed to determine whether school-aged children's use of conversational devices with their friend is sensitive to contextual demands, and how the use of these devices relate to children's perceptions of friendship quality. Overall, we replicated some patterns from the adult literature in our sample of school-aged children, such as higher levels of lexical alignment in TOCs, and more backchannels, higher rates of syntactic alignment, and higher levels of semantic alignment in SOC. We also identified new context-specific patterns, such as greater lexical alignment rate in TOCs and more discourse markers in SOC. We also found that backchannels and semantic alignment, in particular, were strongly related to children's perceptions of friendship quality.

Having replicated a similar pattern as Dideriksen and colleagues (2023), we put forth further evidence that different contexts afford diverse uses of conversational devices, even for children. Like adults, children may be lexically and syntactically aligning to be precise and faster in achieving shared goals in the context of TOCs, and using backchannels and aligning semantically, to maintain social relations in SOC. We also further expanded Dideriksen and colleagues' (2023) study, as we identified new context-specific patterns, such as children lexically aligning at greater rates in TOCs and using more discourse markers in SOC. One possible explanation for this could stem from our participants knowing one another. While Dideriksen and colleagues (2023) paired adult strangers with one another, the current study invited dyads of friends to complete the tasks. It is possible that because friends are familiar with one another, there is less need for more of their turns to be lexically aligned in the context of SOC because they already know one another. Further, we found that the child-to-child discourse generally yielded greater uses of particular devices than previously reported in the context of child-adult discourse (Fusaroli et al., 2023a, 2023b). For example, children lexically aligned with their friends in roughly 33% of their turns across the entire protocol, whereas children lexically aligned to their caregiver in only 23% of their turns (Fusaroli et al., 2023a). In the case of child-to-child discourse, both children share authority and control over the discourse. Rather than relying on an adult, children in child-to-child discourse have free reign over their own linguistic choices, yielding greater uses of these devices than previously reported in the context of child-adult discourse.

Beyond examining whether school-aged children's use of conversational devices with their friend is sensitive to contextual demands, we were also interested in the relationship between language and children's perceptions of their friendship quality. We found that backchannels and semantic alignment were positively associated with positive features of friendship quality, such as companionship and closeness. This corroborates previous literature that suggests that when people are not engaged (e.g., addressee does not backchannel) or change the subject (i.e., not semantically aligning), they feel less close to their interlocutor (Rose et al., 2016).

In sum, the current study has found that children adjust their uses of conversational devices depending on the context. Additionally, our findings suggest that language is a viable tool for studying social relations in developmental studies. Future studies should continue to explore the relationship between language and social relations in developmental samples to build more successful relationships.

References

- Bauminger, Nirit, Solomon, Marjorie, Aviezer, Anat, Heung, Kelly, Gazit, Lilach, Brown, John, & Rogers, Sally J. (2008). Children with Autism and Their Friends: A Multidimensional Study of Friendship in High-Functioning Autism Spectrum Disorder. *Journal of Abnormal Child Psychology*, 36, 135–150. <https://doi.org/10.1007/s10802-007-9156-x>
- Brennan, Susan E., & Clark, Herbert H. (1996). Conceptual pacts and lexical choice in conversation. *Journal of experimental psychology: Learning, memory, and cognition*, 22(6), 1482. <https://doi.org/10.1037/0278-7393.22.6.1482>
- Brinton, Laurel J. (2010). Discourse Markers. *Historical pragmatics*, 285–314. <https://doi.org/10.1515/9783110214284.5.285>
- Buhrmester, Duane & Furman, Wyndol (2008). *The Network of Relationships Inventory: Relationship Qualities Version*. Unpublished measure, University of Texas at Dallas.
- Bürkner, Paul C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of statistical software*, 80, 1–28. <https://doi.org/10.18637/jss.v080.i01>
- Carletta, Jean, Isard, Amy, Isard, Stephen, Kowtko, Jacqueline C., Doherty-Sneddon, Gwyneth, & Anderson, Anne H. (1997). The reliability of a dialogue structure coding scheme. *Computational Linguistics*, 23, 13–31. <http://dl.acm.org/citation.cfm?id=972686>
- Datavyu Team (2014). *Datavyu: A Video Coding Tool*. *Databrary Project*, New York. University. URL <http://datavyu.org>.
- Dideriksen, Christina, Christiansen, Morten H., Dingemanse, Mark, Højmark-Bertelsen, Malte, Johansson, Christer, Tylén, Kristian, Fusaroli, Riccardo (2023). Quantifying the interplay of conversational devices in building mutual understanding. *Journal of Experimental Psychology: General*, 152(3), 864. <https://doi.org/10.1037/xge0001301>
- Duran, Nicholas. D., Paxton, Alexandra, & Fusaroli, Riccardo (2019). ALIGN: Analyzing linguistic interactions with generalizable techNiques—A Python library. *Psychological methods*, 24(4), 419. <https://doi.org/10.1037/met0000206>
- Fusaroli, Riccardo, Bahrami, Bahador, Olsen, Karsten, Roepstorff, Andreas, Rees, Geraint, Frith, Chris, & Tylén, Kristian (2012). Coming to terms: Quantifying the benefits of linguistic coordination. *Psychological Science*, 23(8), 931–939. <https://doi.org/10.1177/0956797612436816>
- Fusaroli, Riccardo, Weed, Ethan, Rocca, Roberta, Fein, Deborah, & Naigles, Letitia (2023a). Repeat after me? Both children with and without autism commonly align their language with that of their caregivers. *Cognitive Science*, 47(11), e13369. <https://doi.org/10.1111/cogs.13369>
- Fusaroli, Riccardo, Weed, Ethan, Rocca, Roberta, Fein, Deborah, & Naigles, Letitia (2023b). Caregiver linguistic alignment to autistic and typically developing children: A natural language processing approach illuminates the interactive components of language development. *Cognition*, 236, 105422. <https://doi.org/10.1016/j.cognition.2023.105422>

- Garrod, Simon, & Pickering, Martin J. (2009). Joint action, interactive alignment, and dialog. *Topics in Cognitive Science*, 1(2), 292-304.
<https://doi.org/10.1111/j.1756-8765.2009.01020.x>
- Hoff, Erika (2010). Context effects on young children's language use: The influence of conversational setting and partner. *First language*, 30(3-4), 461-472.
<https://doi.org/10.1177/0142723710370525>
- Ireland, Molly E., Slatcher, Richard B., Eastwick, Paul W., Scissors, Lauren E., Finkel, Eli J., & Pennebaker, James W. (2011). Language style matching predicts relationship initiation and stability. *Psychological science*, 22(1), 39-44.
<https://doi.org/10.1177/0956797610392928>
- Jucker, Andreas H. (1993). The discourse marker well: A relevance-theoretical account. *Journal of pragmatics*, 19(5), 435-452.
[https://doi.org/10.1016/0378-2166\(93\)90004-9](https://doi.org/10.1016/0378-2166(93)90004-9)
- MacWhinney, Brian (2000). *The CHILDES project: Tools for analyzing talk: Transcription format and programs (3rd ed.)*. Lawrence Erlbaum Associates Publishers.
<https://doi.org/10.1177/026565909200800211>
- Odato, Christopher V. (2013). The development of children's use of discourse like in peer interaction. *American speech*, 88(2), 117-143.
<https://doi.org/10.1215/00031283-2346825>
- Peterson, Carole (1990). The who, when and where of early narratives. *Journal of child language*, 17(2), 433-455. <https://doi.org/10.1017/S0305000900013854>
- Rifai, Olivia M., Fletcher-Watson, Sue, Jiménez-Sánchez, Lorena, & Crompton, Catherine J. (2022). Investigating markers of rapport in autistic and nonautistic interactions. *Autism in Adulthood*, 4(1), 3-11. <https://doi.org/10.1089/aut.2021.0017>
- Rose, Amanda J., Smith, Rhiannon L., Glick, Gary C., & Schwartz-Mette, Rebecca A. (2016). Girls' and boys' problem talk: Implications for emotional closeness in friendships. *Developmental psychology*, 52(4), 629.
<https://doi.org/10.1037/dev0000096>
- Schiffrin, Deborah. (1987). *Discourse markers (No. 5)*. Cambridge University Press.
- Semel, Elisabeth M., Wiig, Eleanor H., & Secord, Wayne A. (1987). *Clinical evaluation of language fundamentals—Fifth Edition (CELF-5)*. Bloomington, MN: NCS Pearson.
- Skarabela, Barbora, Cuthbert, Nora, Rees, Alice, Rohde, Hannah, & Rabagliati, Hugh (2023). Learning dimensions of meaning: Children's acquisition of but. *Cognitive Psychology*, 147, 101597. <https://doi.org/10.1016/j.cogpsych.2023.101597>
- Sprott, Richard A. (1992). Children's use of discourse markers in disputes: Form-function relations and discourse in child language. *Discourse Processes*, 15(4), 423-439.
<https://doi.org/10.1080/01638539209544821>
- Tickle-Degnen, Linda, & Rosenthal, Robert (1990). The nature of rapport and its nonverbal correlates. *Psychological inquiry*, 1(4), 285-293.
https://doi.org/10.1207/s15327965pli0104_1
- Tylén, Krisitan, Fusaroli, Riccardo, Østergaard, Sara M., Smith, Pernille, & Arnoldi, Jakob (2020). The social route to abstraction: Interaction and diversity enhance rule-formation and transfer in a categorization task. *Psyarxiv*.
<https://doi.org/10.31234/osf.io/qs253>
- Yngve, Victor (1970). On getting a word in edgewise. In *Papers from the Sixth Regional Meeting of the Chicago Linguistic Society*, p. 567-578. University of Chicago, Dept. of Linguistics, Chicago.

Proceedings of the 49th annual Boston University Conference on Language Development

edited by Aditya Yedetore,
Rebecca Dufie Bonney, and Yuanyuan Zhang

Cascadilla Press Somerville, MA 2025

Copyright information

Proceedings of the 49th annual Boston University Conference on Language Development
© 2025 Cascadilla Press. All rights reserved

Copyright notices are located at the bottom of the first page of each paper.
Reprints for course packs can be authorized by Cascadilla Press.

ISSN 1080-692X
ISBN 978-1-57473-037-1 (2 volume set, paperback)

Ordering information

To order a copy of the proceedings or to place a standing order, contact:

Cascadilla Press, P.O. Box 440355, Somerville, MA 02144, USA
phone: 1-617-776-2370, sales@cascadilla.com, www.cascadilla.com